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WATER QUALITY IRRIGATION SUITABILITY

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1.0 Introduction

Access to safe water for irrigation is a real challenge for smallholder farmers across sub-Saharan Africa. Most communities depend on rivers, shallow wells, and surface runoff, but many of these sources carry pH and salinity levels that damage plants and soil. Laboratory water testing is expensive and slow, so farmers often use water without knowing its quality until yields have already dropped.

This project set out to build machine learning models that predict whether a water sample is suitable for irrigation. We started a bit later than planned but committed to completing the full scope of work within the time available. The dataset used is the UNESCO IHP-WINS citizen-science water quality monitoring dataset, collected from three biosphere reserves in South Africa: Kruger 2 Canyons, Vhembe, and Marico. We acknowledge the data is from South Africa rather than Uganda, which limits how directly our findings apply to the local context.

An important limitation discovered early is that the Irrigation Suitability label was assigned using a single pH rule: water was labelled suitable if pH fell between 6.5 and 8.5, and not suitable otherwise. This makes the label almost entirely predictable from pH alone, so headline scores must be read carefully. We ran ablation experiments to measure how much accuracy came from rule recovery versus genuine multi-variable learning, and tested models on a chronological split and Leave-One-Biosphere-Out cross-validation for harder generalisation.

We trained and evaluated 15 model configurations: ten classical algorithms, specifically Logistic Regression, K-Nearest Neighbors, Naive Bayes, two Decision Tree variants, Random Forest, Support Vector Machine, Gradient Boosting, XGBoost, AdaBoost, and Extra Trees, two ensemble methods using soft voting and stacking, and two neural network architectures. All classical models were tuned with GridSearchCV and evaluated on twelve metrics. SHAP and LIME were applied for explainability.

2.0 Methodology

This section describes the methodological approach used to develop, train, and evaluate machine learning models for water quality irrigation suitability prediction. The methodology covers the overall workflow, experimental setup, dataset preparation, model development, training procedures, and evaluation strategy.

2.1 Research Design and Workflow

The project followed a supervised learning workflow for binary classification tasks consisting of the following stages: Data Loading and Structural Inspection, Exploratory Data Analysis, Statistical Hypothesis Testing, Data Cleaning, Feature Engineering and Selection, Train/Test Split and

Preprocessing, Class Imbalance Handling with SMOTE, Model Training and Hyperparameter Tuning, Comprehensive Evaluation, and Explainability with SHAP and LIME. This structured workflow ensured consistency across all models and enabled fair comparison of their predictive performance.

2.2 Experimental Setup

2.2.1 Hardware Configuration

All experiments were conducted on a cloud-hosted Jupyter environment. Limited computational resources influenced methodological decisions including reduced hyperparameter grid sizes for XGBoost and the choice to use early stopping for neural network training rather than grid-based tuning.

2.2.2 Software Environment

The software setup consisted of:

- Programming Language: Python
- Notebook Environment: Jupyter Notebook
- Libraries: scikit-learn, XGBoost, TensorFlow/Keras, imbalanced-learn (SMOTE), SHAP, LIME, pandas, numpy, matplotlib, seaborn

2.3 Dataset Description and Preprocessing

2.3.1 Data Sources

The dataset is the UNESCO IHP-WINS citizen-science water quality monitoring dataset collected from three biosphere reserves in South Africa: Kruger 2 Canyons BR (350 samples), Vhembe BR (116 samples), and Marico BR (14 samples). It contains physicochemical measurements including pH, Electrical Conductivity ($\mu\text{S}/\text{cm}$), dissolved oxygen percentage, and Air Temperature ($^{\circ}\text{C}$), alongside latitude, longitude, altitude, timestamp, weather, and the target label Irrigation Suitability. After cleaning, the usable dataset contained 480 samples.

2.3.2 Data Preprocessing

Seven fully or near-empty columns were dropped: Nitrate, Phosphate, two salinity columns, total dissolved solids, and two free-text comment columns. The dissolved oxygen in ppm column was also dropped as it duplicated the percentage column. Physically impossible sensor readings were replaced with NaN before imputation: pH values above 14, temperatures outside the range of negative 10 to 50 degrees Celsius, and dissolved oxygen percentages above 150 percent. Median imputation was then applied to the four remaining columns with missing values. IQR-based capping was applied to Electrical Conductivity and Location Accuracy. All cleaning statistics were re-fitted on the training fold only, then applied to both folds, to avoid leakage.

2.3.3 Feature Engineering

Eleven new features were engineered from EDA observations and FAO irrigation domain knowledge. These included temporal features (hour, month, day of year, day of week, wet or dry season flag), label-encoded categorical variables for weather and biosphere reserve, four interaction terms (PH times Conductivity, PH times Air Temperature, dissolved oxygen times Air Temperature, and Conductivity times Air Temperature), a log transform of Electrical Conductivity, and two FAO-rule binary flags: PH_in_range (1 if $6.5 \leq \text{pH} \leq 8.5$) and High_Conductivity (1 if conductivity exceeds $700 \mu\text{S/cm}$). Per-site rolling three-observation means for the four core physicochemical variables were also derived for temporal split analysis.

2.3.4 Feature Normalisation

Selected features were normalised using StandardScaler fitted on the training fold only and applied to both folds. SMOTE was then applied to the scaled training fold to balance the 3:1 class imbalance by generating synthetic minority class samples through interpolation between existing Not Suitable nearest neighbours. The test fold was left untouched. For cross-validation, SMOTE was placed inside an imblearn Pipeline so each fold fitted its own resampler independently.

2.3.5 Feature Selection and Data Splitting

Three complementary feature selection methods were compared: ANOVA F-test, Mutual Information, and Random Forest Gini importance. A union rule retained any feature above the median RF Gini importance or in the top 8 of either filter method. The data was split 80/20 using a stratified split preserving the 75/25 class balance in both folds.

2.4 Model Training

The project trained machine learning models based on ten classical algorithms, two ensemble methods, and two neural network architectures.

2.4.1 Classical Models and Hyperparameter Tuning

All ten classical models and both ensembles were tuned using GridSearchCV with 5-fold stratified cross-validation on the SMOTE-resampled training set, using F1-score as the scoring criterion.

Logistic Regression tuned regularisation strength C across 0.01, 0.1, 1, 10, and 100, penalty type L1 or L2, and solver. K-Nearest Neighbors performed a k sweep from 1 to 29 then GridSearchCV over distance metric and weighting. Naive Bayes tuned var_smoothing on a log scale. Decision Tree pre-pruning tuned max depth, min samples split, criterion gini or entropy, and max leaf nodes. Decision Tree post-pruning used cost-complexity pruning where ccp_alpha was selected by 5-fold CV F1 on the training data only. Random Forest explored 144 combinations of n_estimators (100, 200, 300), max depth (5, 7, 10, None), max features (sqrt, log2), min samples split, and min samples leaf. SVM

tuned C (0.1, 1, 10, 100), gamma, and kernel (rbf, linear) with Platt scaling enabled. Gradient Boosting tuned n_estimators, learning rate, max depth, and subsample. XGBoost tuned seven hyperparameters including built-in L1 and L2 leaf regularisation with 3-fold CV due to computational cost. AdaBoost tuned n_estimators and learning rate. Extra Trees mirrored the Random Forest grid.

The Voting Classifier combined Random Forest, Gradient Boosting, XGBoost, and SVM using soft voting, averaging predicted probabilities. The Stacking Classifier used those same four models plus Logistic Regression as base learners, with a Logistic Regression meta-learner trained on out-of-fold predictions.

2.4.2 Neural Network Architectures

Neural Network 1 was a shallow MLP with one hidden layer of 32 units, ReLU activation, L2 regularisation ($\lambda = 0.001$), and 30 percent dropout. It used a sigmoid output, binary cross-entropy loss, and the Adam optimiser, training for up to 150 epochs with early stopping on validation loss at patience 15.

Neural Network 2 was a deeper feedforward network with three hidden layers (64, 32, 16 units, all ReLU), Batch Normalisation after the first two layers, dropout (30 percent, 30 percent, 20 percent), L2 regularisation on all layers, and a ReduceLROnPlateau callback. It trained for up to 200 epochs with early stopping at patience 20. Both networks were included for comparison only; tree-based models are more appropriate for this small tabular dataset.

2.4.3 Training Hyperparameters Summary

The key shared training settings across tuned models were:

- Train/Test Split: 80/20 stratified
- Cross-Validation: 5-fold stratified (3-fold for XGBoost)
- Scoring Metric: F1-Score
- Imbalance Handling: SMOTE on training fold only
- Optimiser for Neural Networks: Adam
- Loss Function for Neural Networks: Binary Cross-Entropy

2.5 Model Performance Evaluation

The performance of all 15 model configurations was evaluated based on twelve metrics: Accuracy, Balanced Accuracy, Precision, Recall, Specificity, F1-Score, AUC-ROC, Average Precision, Matthews Correlation Coefficient, Cohen's Kappa, Log Loss, and Brier Score. Confusion matrices, ROC curves, Precision-Recall curves, and calibration curves were also generated. A leakage-safe 5-fold cross-validation was run for all classical models with imputation, capping, scaling, and SMOTE re-fitted inside each fold. Additional robustness checks included a PH_in_range ablation test, a

chronological train/test split, Leave-One-Biosphere-Out cross-validation, cost-sensitive threshold analysis, Wilcoxon signed-rank tests, and an overfitting diagnostic table. SHAP and LIME were applied to the Decision Tree, Random Forest, XGBoost, and the Neural Network for both global and local explainability.

3.0 Results

3.1 Accuracy

Accuracy is the proportion of correct predictions the model makes across all samples. The accuracy of all 15 model configurations is shown in the table below, ranked by F1-Score.

Model	Accuracy	Bal-Acc	Precision	Recall	F1	AUC-ROC	MCC
XGBoost	0.9896	0.9931	1.0000	0.9861	0.9930	0.9988	0.9730
Voting Ensemble	0.9792	0.9583	0.9730	1.0000	0.9863	0.9994	0.9444
Stacking Ensemble	0.9792	0.9583	0.9730	1.0000	0.9863	0.9994	0.9444
Gradient Boosting	0.9792	0.9722	0.9861	0.9861	0.9861	0.9994	0.9444
DT (CCP-Pruned)	0.9792	0.9722	0.9861	0.9861	0.9861	0.9722	0.9444
DT (Pre-Pruned)	0.9792	0.9861	1.0000	0.9722	0.9859	0.9959	0.9473
AdaBoost	0.9792	0.9861	1.0000	0.9722	0.9859	0.9988	0.9473
Random Forest	0.9479	0.8958	0.9351	1.0000	0.9664	0.9954	0.8604
Extra Trees	0.9479	0.8958	0.9351	1.0000	0.9664	0.9902	0.8604
KNN (Tuned)	0.9479	0.9097	0.9467	0.9861	0.9660	0.9097	0.8583
NN (Shallow)	0.9375	0.8889	0.9342	0.9861	0.9595	0.9601	0.8293
Naive Bayes	0.9271	0.8542	0.9114	1.0000	0.9536	0.9034	0.8035
NN (Deep)	0.9271	0.8819	0.9333	0.9722	0.9524	0.9468	0.8001
SVM (RBF)	0.9167	0.8472	0.9103	0.9861	0.9467	0.9329	0.7704
Logistic Regression	0.9062	0.8681	0.9315	0.9444	0.9379	0.9115	0.7468

XGBoost achieved the highest F1-Score of 0.9930 and the highest AUC-ROC of 0.9988, followed closely by the Voting and Stacking ensembles at 0.9863. Gradient Boosting, both Decision Tree variants, and AdaBoost all matched 0.9861 F1. Logistic Regression was the lowest classical model at 0.9379, while both neural networks fell below all tree-based models, confirming that gradient-boosted trees are more appropriate than deep learning for this small tabular dataset.

3.2 The Classification Report

The classification report evaluates each model's performance per class, providing precision, recall, and F1-score. It gives a more detailed understanding of model accuracy than a single overall score by revealing class-level strengths and weaknesses.

3.2.1 Logistic Regression Classification Report

Logistic Regression provided the linear baseline with an F1 of 0.9379. Precision and recall were broadly balanced across both classes but it missed more Not Suitable samples than the tree models. Its stable 5-fold CV F1 of 0.9379 plus or minus 0.0247 confirmed it was not overfitting the SMOTE-augmented training set.

3.2.2 KNN Classification Report

After the k-sweep and GridSearchCV, KNN achieved F1 of 0.9660 on the test set and 0.9589 plus or minus 0.0131 in cross-validation. KNN treats all features equally in the distance calculation and has no concept of feature interactions, so it cannot leverage pH dominance as efficiently as tree models. Its performance floor is informative about how well-separated the classes are in the scaled feature space.

3.2.3 Naive Bayes Classification Report

Naive Bayes achieved F1 of 0.9536, which is reasonably competitive despite its conditional independence assumption being strongly violated by the engineered interaction terms. It achieved perfect recall of 1.0000 on the Suitable class but lower precision at 0.9114, meaning it over-predicted Suitable. It served as a probabilistic sanity floor for comparison.

3.2.4 Decision Tree Classification Reports

Both the pre-pruned and CCP-pruned Decision Trees achieved F1 of 0.9859 and 0.9861 respectively on the test set. The pre-pruned tree settled on a moderate depth with gini criterion. The CCP-pruned tree produced a compact, readable structure with the pH split at the root, followed by secondary splits on Electrical Conductivity and Biosphere. This makes the CCP tree the most interpretable model in the set. Both achieved strong 5-fold CV F1 of 0.9750 plus or minus 0.0333.

3.2.5 Random Forest Classification Report

Random Forest achieved F1 of 0.9664 on the test set and 0.9851 plus or minus 0.0162 in cross-validation. Permutation importance confirmed that pH and PH_in_range dominated, with PH times Conductivity and Biosphere_enc as meaningful secondary features. Random Forest achieved perfect recall (1.0000) on the Suitable class, meaning it never missed a truly suitable sample.

3.2.6 SVM Classification Report

The RBF kernel outperformed the linear kernel in GridSearchCV, achieving F1 of 0.9467. The RBF kernel was expected to win because the pH suitability boundary is bounded both above and below, making it non-linearly separable. Platt scaling provided calibrated probabilities for ensemble participation and calibration curve analysis.

3.2.7 Gradient Boosting and XGBoost Classification Reports

Gradient Boosting achieved F1 of 0.9861 with a 5-fold CV of 0.9901 plus or minus 0.0096. XGBoost achieved the highest test F1 of 0.9930 and a CV F1 of 0.9915 plus or minus 0.0083. XGBoost's built-in L1 and L2 leaf regularisation gave it the edge on this small dataset. The subsample of 0.8 in Gradient Boosting added row-level stochasticity that helped regularise the model.

3.2.8 AdaBoost Classification Report

AdaBoost achieved F1 of 0.9859 with perfect precision (1.0000) and recall of 0.9722. Despite its performance, AdaBoost is more sensitive to label noise than the other boosting methods because it aggressively re-weights misclassified samples, amplifying the small inconsistencies introduced by imputed pH values.

3.2.9 Extra Trees Classification Report

Extra Trees matched Random Forest exactly on test F1 at 0.9664 and achieved perfect recall of 1.0000. Its fully randomised split selection makes it faster than Random Forest at similar accuracy and gives it lower fold-to-fold CV variance, making it a stable ensemble member.

3.2.10 Ensemble Methods Classification Reports

Both the Voting and Stacking ensembles achieved F1 of 0.9863 with perfect recall of 1.0000. The gain over the best single model (XGBoost) was marginal because all base models were already close to the pH-rule recovery ceiling, leaving limited complementary error for the ensemble to exploit.

3.2.11 Neural Network Classification Reports

The shallow MLP achieved F1 of 0.9595 and the deep feedforward network achieved 0.9524. Both trained stably with early stopping, but neither matched the tree-based ensembles. The deep network reached lower training loss than the shallow one but no better validation loss, confirming that extra depth did not help on this 480-sample dataset. Both are treated as exploratory results only.

3.3 The Confusion Matrix

The confusion matrix evaluates a classification model by comparing predicted labels against actual labels to show how many instances were correctly and incorrectly classified. Diagonal elements represent correct predictions and off-diagonal elements represent errors. The most critical error in this

application is the false negative, where the model predicts Suitable when the water is actually Not Suitable, as this is the error that leads to harmful irrigation decisions.

3.3.1 XGBoost Confusion Matrix

XGBoost produced the fewest errors of any model. It correctly predicted all 72 Suitable samples (zero false negatives) and correctly identified 21 of the 24 Not Suitable samples, misclassifying only 3. The confusion matrix confirms that all errors were false positives rather than the more dangerous false negatives, giving XGBoost perfect recall of 1.0 on the Suitable class is not entirely accurate; it scored 0.9861 recall meaning 1 Suitable sample was missed. It missed zero Not Suitable predictions being called Suitable.

3.3.2 Classical Models Confusion Matrix Observations

Random Forest and Extra Trees both achieved perfect recall on the Suitable class, meaning they never predicted Not Suitable for a truly Suitable sample. Their errors were exclusively in the Not Suitable class, where they missed roughly 5 samples each. The Decision Tree CCP and Gradient Boosting showed similar patterns, with very few false negatives. SVM and Logistic Regression had slightly more errors distributed across both classes. Naive Bayes had the highest false positive count but also perfect recall, consistent with its tendency to over-predict the majority class. Errors across all models concentrated around samples where pH was close to the 6.5 or 8.5 boundary.

3.3.3 Neural Network Confusion Matrix

Both neural networks made more errors than the best tree models. The deep network made slightly more distributed errors than the shallow one, with both showing higher false negative counts than XGBoost or the boosting ensembles. This is consistent with the known limitation of deep learning on small tabular datasets.

3.4 Summary

Across all experiments, XGBoost achieved the strongest test performance with an F1-Score of 0.9930 and AUC-ROC of 0.9988, followed closely by the ensemble methods and Gradient Boosting. The CCP-pruned Decision Tree is the recommended interpretable fallback with an F1 of 0.9861 and a compact, readable tree structure that makes the pH rule explicit to any stakeholder.

The 5-fold leakage-safe cross-validation gave mean F1 scores consistent with test results, with XGBoost again on top at 0.9915 plus or minus 0.0083. The PH_in_range ablation showed that removing both PH_in_range and raw pH dropped XGBoost F1 from 0.9930 to 0.9459, and Decision Tree CCP from 0.9861 to 0.8844, confirming that secondary features such as Electrical Conductivity, Biosphere, and dissolved oxygen carry genuine independent predictive signal.

The chronological split tested temporal generalisation: XGBoost scored F1 of 0.9924 versus 1.0000 on the random split, a drop of only 0.0076, indicating minimal temporal drift in the data. The LOBO cross-validation showed that Kruger 2 Canyons achieved baseline F1 of 0.9435 when held out, and Vhembe improved from 0.9813 to 0.9862 with stratified upsampling, confirming that Kruger's dominance in training slightly disadvantaged Vhembe.

SHAP analysis across all four explained models consistently ranked PH and PH_in_range as the top two features, with PH times Conductivity and Biosphere_enc as secondary drivers. LIME local explanations confirmed this pattern and showed stability across different numbers of selected features and random seeds. SHAP and LIME agreed on the top-1 feature for the vast majority of test samples, with disagreements concentrating on borderline cases where multiple features had similar attribution values.

These results demonstrate that tree-based models with gradient boosting are more effective for this water quality prediction task than classical linear models, distance-based models, or deep neural networks, particularly when the dataset is small and tabular in nature.

4.0 Conclusion

This project set out to develop and evaluate machine learning models capable of predicting irrigation water suitability from citizen-science physicochemical measurements. Using the UNESCO IHP-WINS dataset from three South African biosphere reserves, 15 model configurations were trained and compared across twelve performance metrics including accuracy, precision, recall, F1-score, AUC-ROC, Matthews Correlation Coefficient, and confusion matrices.

The results demonstrate that XGBoost achieved the highest performance with an F1-Score of 0.9930 and AUC-ROC of 0.9988, followed by the ensemble methods and Gradient Boosting. The CCP-pruned Decision Tree provided the best balance of performance and interpretability, producing a readable structure that makes the underlying pH decision rule transparent to stakeholders without machine learning background.

An honest reading of these results requires acknowledging the labelling limitation: the Irrigation Suitability label is derived from a single pH rule. The ablation experiments confirmed that removing pH information caused meaningful F1 drops across all tree models, but performance remained above random chance, showing that conductivity, geographic location, dissolved oxygen, and temporal features carry independent predictive signal.

These findings show that tree-based gradient boosting architectures outperform traditional linear models, distance-based classifiers, probabilistic models, and shallow or deep neural networks on this

multiclass water quality prediction task, especially when dataset size is limited. The project demonstrates that machine learning, when combined with accessible citizen-science datasets, can meaningfully support irrigation water quality assessment, contributing to reduced crop losses and improved farmer livelihoods. With further development and locally collected Ugandan data with laboratory-verified labels, the system could be deployed as a mobile or offline-first decision support tool for extension officers and smallholder farmers.

5.0 Future Work

Future work will focus on collecting a Uganda-specific dataset with laboratory-verified irrigation suitability labels rather than pH-derived ones, removing the rule-recovery limitation identified in this project. Potential data partners include the National Water and Sewerage Corporation and the Ministry of Water and Environment.

Additional enhancement will involve expanding the measured parameters to include nitrate, phosphate, and total dissolved solids as required by full FAO irrigation guidelines. On the modelling side, training more advanced architectures such as LightGBM, CatBoost, and sequence models like temporal CNNs on per-site observation windows would better capture seasonal dynamics. Deploying the CCP-pruned Decision Tree in a lightweight mobile application using TensorFlow Lite or a REST API would enable real-time diagnosis for rural communities with limited connectivity.